DYLAN DUTTON

Astrophysicist and Software Engineer

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I am a physics PhD candidate at the University of North Carolina at Chapel Hill as well as a researcher and software engineer for the global robotic telescope network - Skynet. As a software engineer, I specialize in back-end design and development focusing on the automation of Skynet's robotic telescopes. As a researcher, I study gravitational-wave events to get a definitive measure of the expansion rate of the universe.

EDUCATION

University of North Carolina at Chapel Hill	Physics	PhD (Ongoing)
University of North Carolina at Chapel Hill	Physics	M.S. 2021
University of North Carolina at Chapel Hill	Physics	B.S. 2018

WORK EXPERIENCE

Software Engineer

Skynet Robotic Telescope Network | 2018 -2019

- Led development on an autonomous algorithm that controls dozens of telescopes simultaneously
- Collaborated with the end-users to develop an API and UI to monitor the telescopes' observations
- Authored a peer-reviewed systems paper on the algorithm

Senior Physics Instructional Assistant

University of North Carolina at Chapel Hill | 2015 - 2018

- Diagnosed, fixed, and tested physics lab equipment for five physics courses
- Conducted experiments to produce the expected initial results

PROJECTS

The Campaign Manager

Skynet Robotic Telescope Network

The Campaign Manager (CM) autonomously scales and schedules exposures on dozens of Skynet's robotic telescopes simultaneously. The CM is useful for follow up to various transient phenomena and is available to 10,000s of Skynet users. I wrote the core algorithm of the CM in Python3, the RESTful API using Python3 and JSON, the webpages using HTML/CSS, JavaScript, AJAX, and JQuery, and the interactions with Skynet's relational database using SQL and SQLAlchemy.

Radio Image-Processing Algorithm

Skynet Robotic Telescope Network

After starting on the radio image-processing algorithm in 2015, I became the lead developer in 2017. The algorithm leverages a new outlier rejection method (that I also helped to develop) called Robust Chauvenet Rejection (RCR) to clean highly contaminated (up to 80%) radio telescope data. The algorithm is written in C++ to make use of low-level memory management.

SKILLS

Technical (Proficient)

Python, C++, C#, JavaScript, TypeScript

Technical (Familiar)

Vue, React, JQuery, SQL, Mathematica, JSON

Technology and Apps

Git, BitBucket, VSCode, Node/npm, Visual Studio, LaTeX, Unity, Blender, Photoshop, XD, Excel, AJAX, Arduino, Linux

Writing

4 Refereed Publications
5 Authored Grants

Featured Courses

Math Methods of Physics
Computational Modeling
Differential Equations
Calculus I & II & III
Linear Algebra
Discrete Methods
Numerical Techniques

LANGUAGES

English (Native)
Arabic (Conversant)

github.com/astrodyl
linkedin.com/in/astrodyl

Open-Source Gravitational Wave Plotting Tool

The University of North Carolina at Chapel Hill

I am leading the development on the gravitational wave (GW) plotting tool which visualizes GW signals and educates users about the relationship between the physical parameters of GW events and the detected signal. The tool will be integrated into 20+ college curricula and will be used by 1000s of students every semester. The end-user web-interface is written in HTML/CSS and TypeScript. The back-end, which handles data manipulation, is written in Python3.

RESEARCH EXPERIENCE

Multi-Messenger Measurement of the Hubble Constant

The University of North Carolina at Chapel Hill | 2019 - Present

- Leading a multi-institution, global observing campaign of GW events
- Measuring viewing angles using semi-numerical and analytical models
- Developing open-source web tools for parameter measurements

Radio Image Photometry

The University of North Carolina at Chapel Hill | 2015 - 2018

- Led a team of three on a single radio image photometry algorithm
- Integrating algorithm into Skynet to automatically process 1000s of radio observations per year
- Published a peer-reviewed paper on the algorithm

Harsh-Environment Electronics

Elizabeth City State University | 2014 - 2015

- Fabricated and characterized micro-electronic contacts on wide-band semiconductors for applications in extremely harsh environments (high temperature and high radiation)

TEACHING EXPERIENCE

Head Astronomy TA

The University of North Carolina at Chapel Hill | 2018 - Present

- Leading a group of four introductory astronomy TAs
- Teach multiple sections of up to 35 students each
- Developing grading and data visualization software

PUBLICATIONS

- D.A. Dutton, D.E. Reichart, J. Haislip, and et. al. Skynet's New Observing Mode: The Campaign Manager. PASP, 134:1031, Jan. 2022.
- J.R. Martin, D.E. Reichart, D.A. Dutton, and et. al. Skynet Algorithm for Single-dish Radio Mapping. I. Contaminant-cleaning, Mapping, and Photometering Small-scale Structures. ApJS, 240(1):12, Jan 2019.
- M.P. Maples, D.E. Reichart, and et. al. including D.A. Dutton. Robust Chauvenet Outlier Rejection. ApJS, 238:2, August 2018.
- A.S. Trotter, D.E. Reichart, and et. al. including D.A. Dutton. The Fading of Cassiopeia A, and Improved Models for the Absolute Spectrum of Primary Radio Calibration Sources. MNRAS, 469:1299-1313, August 2017.